A conceptual history of the emergence of bounded rationality

Matthias Klaes, Keele University, office@mklaes.net
Esther-Mirjam Sent, University of Notre Dame and Netherlands Institute for Advanced Study, sent.2@nd.edu

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Abstract

Whereas multiple interpretations of the concept of bounded rationality are currently enthusiastically employed by a wide range of academics, their colleagues of one hundred years ago did not have this concept available, but made use of related notions. This paper traces the history of the emergence of 'bounded rationality' from the first appearance of 'limited intelligence' in 1840 and 'finite intelligence' in 1880 through related concepts such as 'incomplete', 'limited', 'administrative', and 'approximate' rationality during the first half of the twentieth century to its inception in 1957 at the hands of Herbert Simon. In the process, it locates the semantic field from which the notion of bounded rationality emerged and attends to the co-emergence of terms such as 'limited rationality'. This allows us to analyse the process of metaphorical crystallisation in which 'bounded rationality' assumed the dominant position within its semantic field.

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"That's a great deal to make one word mean," Alice said in a thoughtful tone.
"When I make a word do a lot of work like that," said Humpty Dumpty, "I always pay it extra."

Lewis Carroll, *Through the Looking Glass*

1. Introduction

In 1992, Herbert Simon noted that "[r]eaders would not be deceived by the claim that economists flocked to the banner of satisficing man with his bounded rationality. The 'flocking' was for a long time a trickle that is now swelling into a respectable stream" (Simon 1992: 266).[1] For instance, Thomas Sargent published *Bounded Rationality in Macroeconomics* in 1993 and tried to make connections to Simon's bounded rationality contributions: "Herbert Simon and other advocates of 'bounded rationality' propose to create theories with behavioral foundations by eliminating the asymmetry that rational expectations builds in between the agents in the model and the econometrician who is estimating it" (Sargent 1993: 21-22). However, whereas Sargent embraced neoclassical theory and parallel adaptive computing systems, Simon dismissed neoclassical theory in favour of behavioural approaches and rejected parallel systems in support of the serial symbol processing approach (Sent 1997).

Bounded rationality has recently made its way into not only rational expectations approaches, but also game theoretic contributions. Yet, in the paper that stimulated research on modelling bounded rationality through automata, game theorist Robert Aumann traced his suggestion back no further than Roy Radner's insights. In particular, Aumann (1981) claimed that "[f]inite memory has some conceptual ties to Radner's bounded rationality" (21). On another occasion, Aumann (1986) examined how bounded rationality approaches "have evolved over the past ten or fifteen years" (5). Only later did Aumann (1997) make a connection between his suggestion and Simon's work: "To my knowledge, this area was first extensively investigated by Herbert Simon" (3). These efforts give rise to two paradoxical observations. First, whereas Simon saw bounded rationality as part of an alternative research programme, game theorists employ his ideas to further the prevailing orthodoxy. Second, in adopting bounded rationality, game theorists find themselves in the position of using bounded rationality to define rationality (Sent forthcoming).

Historians of economics might respond to the efforts of Sargent and his fellow rational expectations economists and of Aumann and his game theory colleagues by illustrating that the recent appropriations hinge on an unsuitable reading of Simon's insights, or by applauding the progress to which Simon's earlier contributions have been subject. In contrast to such evaluations, we propose to study the history of 'bounded rationality' — 'BR' from now on — without imposing

[1] Just one year earlier, Simon (1991) had complained: "My economist friends have long since given up on me, consigning me to psychology or some other distant wasteland" (385).

our own or anyone else's reading on it. What we are interested in is the origin and diffusion of the concept, which we understand naturalistically as an entity embedded in a network of connections that evolve over time. In short, we propose to study BR as an institution. More specifically, we focus here on its institutionalisation. We are not interested in whether Sargent or Aumann 'got it right'. Rather, we are fascinated by the implicit assumption in such an evaluation that there is a 'thing' that has a proper reading and an improper interpretation. In 1896, such an assessment would have been meaningless. One hundred years later, a survey article in the *Journal of Economic Literature* asked "Why Bounded Rationality?" (Conlisk 1996) and developed four answers: evidence, success, methodology, and scarcity. Whereas this overview focused almost exclusively on the last fifteen years (Conlisk 1996: 669), we are interested in exploring the emergence of BR as a concept over a much longer period.

Section 2 discusses our historiography and method. In the subsequent section we analyse the semantic field from which BR has emerged. Section 4 explores Simon's experiments with several concepts derived from the field and attends to the initial diffusion of BR. Finally, section 5 gives a short overview of the subsequent consolidation and diffusion of BR.

2. Conceptual History and Semantic Fields

Where should the history of a concept start? If we study concepts naturalistically as entities emerging and diffusing through the economic literature, what we need to do is locate the texts that employ the concept explicitly. While we may be inclined to talk about related analyses that use different expressions, we should avoid the temptation of anachronistic exegesis that would identify BR in every contribution of the past bearing some similarities to present-day interpretations of the concept. On the other hand, we should recognise that the continuity of scholarly discourse over time reaches beyond the confines of individual expressions. It is thus necessary to strike a balance between sketching the parochial history of a single concept looked at in isolation, and the embarrassment of riches that the traditional historian of ideas is in danger of uncovering from the past. Our answer to this dilemma is a study of the historicity of the semantic field from which concepts emerged as a dominant focal point.

To formulate the idea of a semantic field in more precise terms, let us first further clarify our approach to conceptual history. Contributors to this research area have tended to understand concepts in two different ways (Richter 1996). On the one hand, historians of ideas have favoured a mentalistic interpretation, keeping to Lovejoy's (1936) precept of tracing concepts as implicit assumptions or mental habits through various disciplines and epochs. According to this idealist interpretation, certain terms can be seen as the historical manifestation of concepts, and a concept may be represented by several different expressions in the course of its history and its migration

across various contexts.

On the other hand, and partly as a reaction to the assumption of an underlying continuity of intellectual history in which shared mental content forms the stable unit of analysis, the second strand of conceptual history, associated with the German tradition of *Begriffsgeschichte*, puts more emphasis on the discontinuous elements of historical development. According to this interpretation, concepts must be understood as a certain class of words (Koselleck 1972; Palonen 1985, 1989).[2] Hence, they become a concrete entity the occurrence of which can be traced empirically in historical texts, logically prior to a textual exegesis. This contrasts with the idealist approach, which, by tracing a common meaning in a variety of expressions, celebrates an underlying unity of the intellectual record of the past. Conceptual history of the *Begriffsgeschichte* type is concerned with the diversity of concept application. It considers ambiguity of certain expressions as one of the defining characteristics of their status as social institutions (Koselleck 1972; Klaes 2001), and it is both this status and their systematic — instead of context related — ambiguity that distinguishes them from other words (that are not regarded as concepts).

It is a strength of idealist conceptual history that it looks beyond the maze of the 'penumbra of vagueness' (Popper 1945: 19) of the terms we employ, and that it embraces historical developments reaching beyond the usage pattern of a single expression only. At the same time, it is however precisely this concern that leaves idealist conceptual history vulnerable to anachronistic interpretations of the past. But is it possible for the conceptual historian to apply a criterion of continuity different from shared meaning to consider the history of two or more expressions?

Clearly, as long as we confine ourselves to the level of texts only, some relationship that is ultimately based on semantics is necessary to establish diachronic continuity across a set of expressions that the historian considers related. What we are looking for, however, is a weaker definition of this set than in terms of shared conceptual content. In order to address the historical continuity of concept application across several expressions from a naturalistic perspective, we follow conceptual historians who encourage the study of semantic fields (Hampsher-Monk et al. 1998b: 2). A semantic field in the sense employed here is a collection of emergence and diffusion trajectories of individual concepts (understood as lexemes, cf. fn. 2). In historiographical terms, its

[2] To be more precise in linguistic terms, *Begriffsgeschichte* is concerned with lexemes (Lyons 1977: 18-20). The distinction between token words as they appear in written text, word-forms, expressions, and lexemes is subtle. Consider the word 'find'. Clearly, as a first step, we are abstracting from the physical typeface when we speak of the same word (let alone added complexities due to the difference between written and spoken language, etc). But we would also say that 'find' and 'found' are the same word, yet appearing in different forms. Linguists — though not all of them — tend to refer to this underlying word as a lexeme. This constitutes thus a further layer of abstraction, as all we ever encounter are the forms of this lexeme. When we refer to a lexeme as such, we typically use its citation-form as it can be found in dictionaries, for example. The term 'expression', on the other hand, is used in philosophical semantics, but rarely with a clear distinction between expressions, forms, and lexemes. Our allegiance is to the linguistic interpretation, but for the present paper we need not be concerned with these technical details and will thus use the terms 'word', 'expression', 'notion' and 'concept' interchangeably.

analysis consists of the aggregation of individual conceptual histories into an investigation of the complex interdependencies of the evolving institutional network of concepts.[3] In the present study it is thus the semantic field that lends conceptual history the continuity that idealist conceptual history seeks to locate in the continuity of a single unchanging meaning of a concept.

Intrinsic to the definition of a semantic field, as a set of co-evolving concepts, is some arbitrariness regarding the nature of the interrelatedness of the concepts thus included in the field. Once diversity in concept application is allowed, there is no fast and easy criterion to determine whether a particular expression should be regarded as part of the semantic field or not. This contrasts sharply with the practice of idealist conceptual history, where the question whether a particular concept should be included in the historical account is immediately one of exegesis, and not of definition of the historiographical categories used. If every expression carried only one meaning, the task would be relatively straightforward. A semantic field would then simply consist of all the synonyms of a particular concept.

Conceptual history in the tradition of *Begriffsgeschichte* is concerned not with easily definable terms expressing an underlying common idea, but with inherently ambiguous notions. Synonymy of expressions as a criterion for belonging to the same semantic field needs to be relaxed to account for the intrinsic polysemy and ambiguity found in the semantic fields studied by *Begriffsgeschichte*. Pending a more thorough exploration of a principled approach to this issue (Klaes 2002), we employ a heuristic approach in this paper.

Specifically, we adopt a three-step procedure for heuristically arriving at the semantic field of BR by identifying candidate conceptual trajectories. First, a list of related keywords was drawn up that both authors judged to be sufficiently related to form the basis for the semantic field associated with BR.[4] In a second step, this list then served as the search criterion for identifying the records making up our historical corpus. Third, upon reading through the entries, the list was further revised by either eliminating expressions that were found not to be related to the field associated with the concept of bounded rationality (henceforth called the 'BR field'), or by adding new expressions that were found to be so. In the latter case, the respective expressions served as additional search criteria to add further conceptual trajectories to the field.

In our effort to analyse the historical development of BR, we used JSTOR as the database, which contains a mass of information that can be digested sensibly. It archives 242 journals, 66,730 issues,

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[3] In spirit, our analysis encourages to extend the historical lens to the histories of the individuals, groups, professional bodies, and all the other entities that sustain the emerging conceptual network of the semantic field, and figure prominently in institutional history of economics (for a more detailed description of this historiographical stance in the history of economics, see http://eh.net/lists/archives/hes/nov-1998/0011.php).

[4] A more systematic approach (Fortier, Keen and Fortier 1997) would have been to convene an expert panel to draw up the list or to interview members of the various discourse communities employing the BR concept.

734,651 full-length articles, 1,612,462 articles, and 9,900,014 pages in collections covering arts and sciences, business, ecology and botany, and general science. The journals are carefully selected based on the number of institutional subscribers, a citation impact factor analysis, recommendations from experts in the field, and the length of time a journal has been published. The gap between the most recent published issue of a journal and the most recent issue available on JSTOR varies from two to five years. Since JSTOR relies on a careful selection mechanism, we are confident that our findings are a positive indication of the conceptual history of bounded rationality. While they might underestimate their true scale, even within the discipline of economics, we do not consider this to be a damaging constraint. Even though JSTOR contains only journal publications, it does give us insights into manuscripts through book reviews published there. Though our graphical and tabular representations include the reviews and not the books, the latter were checked for some verbal analyses. We further consulted the *Oxford English Dictionary* to learn about the historical origins of certain concepts found in our JSTOR database.

Our heuristic procedure led us to the following keywords: 'administrative rationality', 'approximate rationality', 'bounded intelligence', 'bounded rationality', 'boundedly rational', 'constrained rationality', 'finite intelligence', 'finite rationality', 'incomplete rationality', 'limited intelligence', 'limited rationality', and 'restricted rationality'. For each entry in our database, we noted the sentence in which the concept occurred and further, if deemed appropriate, included one or two sentences directly before the occurrence or one or two sentences immediately after. Our search resulted in 1029 entries. Because of JSTOR's 'moving wall', we eliminated the items from 1996 onwards.

3. The Semantic Field of Bounded Rationality

In this section we report on our analysis of the BR field from 1840 to 1995. Table 1 shows the set of heuristically generated keywords, together with their earliest and total occurrence in the corpus. Several implications can be drawn from this overview. Note first that not all of the keywords figure equally prominently in the corpus. 'Bounded intelligence' and 'restricted rationality' in particular can be safely excluded from further consideration of the semantic field of BR.[5] Secondly, the appearance of BR in 1957 allows us to identify the semantic field from which the concept has emerged. This centres on the notions of 'limited intelligence', 'finite intelligence', together with 'limited rationality', and 'approximate rationality'.[6]

[5] We would not like to rule out that those expressions might join the field in the future, but do not attach any particular further significance to the fact that we considered them as potential candidates ex ante.

[6] We decided to exclude 'administrative rationality' from the field, due to its being largely interpreted in terms of the bureaucratic rationale of an organisation (e.g. "Efficacy, efficiency, and profitability in the narrowest economic sense have become the only standards of administrative rationality." Berger et al 1969: 454), and due to the absence

Keyword Heuristic											Total
Limited Intelligence	1840										73
Finite Intelligence		1880									40
Incomplete Rationality			1922								7
Limited Rationality				1945							105
Administrative Rationality					1947						48
Approximate Rationality						1948					9
Bounded Rationality							1957				626
Finite Rationality								1972			17
Constrained Rationality									1978		7
Restricted Rationality										1983	1
Bounded Intelligence											0

Table 1: The Emergence of the Semantic Field of BR

Not all of these notions experience the same fate, though (Figure 1). ‘Finite intelligence’, the most prominent expression between 1880 and 1935, leads a marginal existence in the field thereafter, while ‘limited intelligence’ exhibits a consistent presence throughout the period under consideration. The pattern of ‘limited rationality’, ‘administrative rationality’ and ‘approximate rationality’ is different. All emerge within a decade after World War II. The first two exhibit growth of similar scale up to 1975, with ‘approximate rationality’ trailing slightly behind. From then on, the ways part. ‘Limited rationality’ continues on a steep trajectory of exponential growth. ‘Administrative rationality’ settles for an overall linear growth pattern, while ‘approximate rationality’ virtually disappears. ‘Incomplete rationality’ has one early ephemeral occurrence, only to re-occur several decades later to join the marginal status of ‘finite intelligence’ and ‘approximate rationality’.

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of any obvious semantic connections to other expressions in the field.

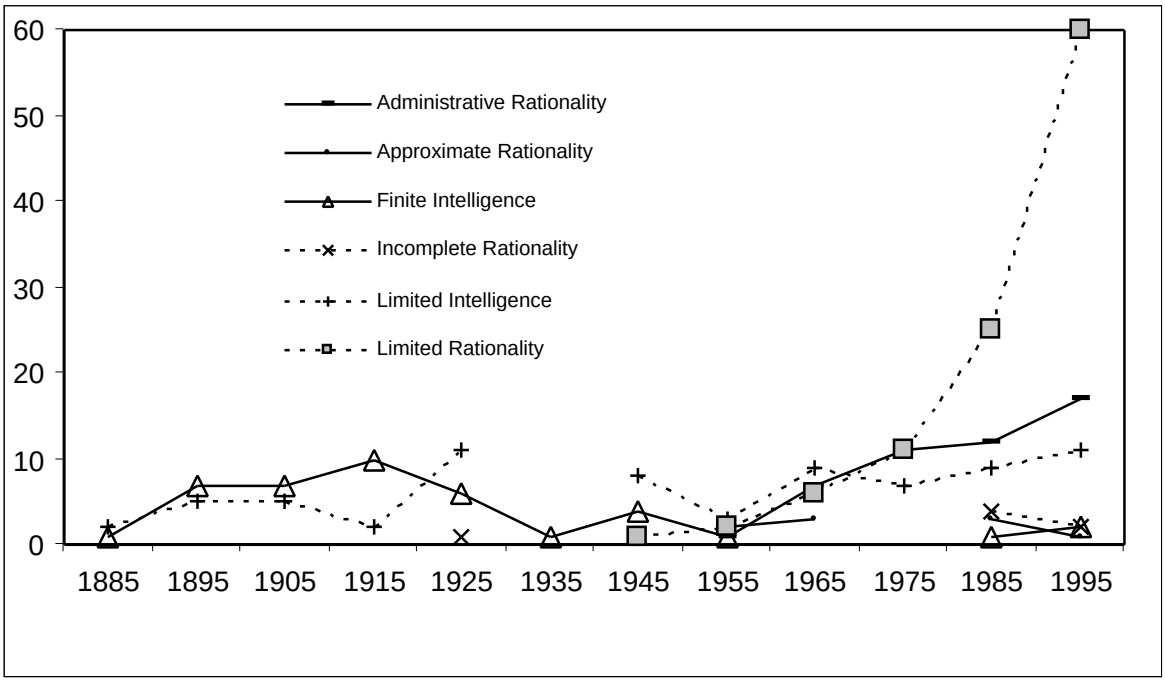


Figure 1: The Semantic Field excluding BR (1876-1995, per annum counts for 10-year intervals)[7]

Above, it was noted that this grouping of expression into a common semantic field has been of a heuristic nature. Hence, not all of the notions that we as present-day observers of the development of BR could plausibly suggest as candidates for the field are in fact realised.

'Limited intelligence' can be identified as a living term of common sense language throughout the period under consideration (1840-1995). Due to the composite nature of this expression, the *Oxford English Dictionary (OED)* does not document its historical emergence. It seems plausible to date it not before the 17th century, this being the time when 'limited' acquired the meaning of 'bounded' or 'restricted'. It occurs first in our corpus in 1840 (Report to the Council of the Statistical Society), followed by more frequent occurrences from 1880 onwards, in a whole range of disciplines, including the sciences (Duke of Argyll 1880: 231), language studies (Menger 1894: 159), politics (Emery 1895: 84) and economics (Adams 1886: 103).[8] A typical usage would be similar to Walker (1894: 244), another economist: "It is, perhaps, easier for a man with limited intelligence to make a selection if the banks have large capital and are of semi-national importance, … ."

An important aspect to note in the context of 'limited intelligence' is the sometimes rather prejudicial, class-conscious and even overtly racist context in which the expression is employed. Fowler (1899: 297) writes that simple love of truth "may coexist with an extremely limited

[7] This figure has not been adjusted for the total number of entries in JSTOR, since these numbers exhibit a roughly linear trend that does not affect the graph too much for our purposes. A normalized figure is available from the authors by request.

[8] Bain (1876: 193) speaks of men whose "intelligence was … limited."

intelligence." Danforth (1925: 185) finds that "[t]he chief difficulty in educating all mankind to the point of practicing eugenics would lie in inducing people with such limited minds as the average man possesses to want to try it." Bryce (1921: 675) revels in noting that "[m]assacre is the expedient to which the Turks always resort; their limited intelligence, incapable of reforming their administration, flies at once to bloodshed." Slichter (1945: 111) comments that while "[m]ost members of the work force are industrious, reliable, reasonably intelligent, and able to get on with other people, but about 4 or 5 percent of the job seekers have very limited intelligence ... ." Logan (1951: 561) finally appraises the historical value of the memoirs of Isaac Jefferson, one of the slaves of Thomas Jefferson, by alerting the reader that "the value of the slave's observations and comments might be lessened by his limited intelligence."

In contrast to the non-technical, common-sense use of 'limited intelligence' that occurs in a wide range of disciplines and is partly subject to very obvious socio-political overtones, 'finite intelligence' is a technical term largely confined to the philosophical literature. The earliest documented occurrence of 'finite intelligence' in the *OED* can be found in John Norris's *Essay Towards the Theory of the Ideal or Intelligible World* (1701: I.i.21), where it is used in contrast to properties ascribed to God. The first source in our corpus in which it is employed dates from the late 19th century (Caird 1880: 550). In this early part of the period under consideration, 'finite intelligence' is typically used as a technical term by philosophers to set up a contrast to "the one universal thought" (Archer-Hind 1889: 131), the "Absolute" and "Infinite" (Denovan 1889: 218), or the "Supreme Being" (Chrysostom 1894: 148). Classical authors frequently referred to in this context are Thomas Aquinas and Spinoza.

'Finite intelligence' was however not exclusively used in a religious context, or in the old philosophical reflection on the nature of the Absolute. For example, a number of late 19th-century sources refer to the German philosopher Immanuel Kant (e.g. Seth 1889; Schurman 1893) who sought to determine the nature and limits of human knowledge and who questioned how faithfully mental images represent physical objects. In Kant's view on epistemology, knowledge begins with experience, which results from the cooperation of one's senses and one's mind. Whereas the senses determine the *matter* of experience, the mind governs the *form*. Whereas the senses control the *content* of experience, the mind influences the *structure*. As a result, we know a phenomenal world, but we have no knowledge of things-in-themselves. In other words, we have no certain knowledge of noumenal reality, of things-in-themselves as they exist apart from the mind. Intelligence, then, is 'finite' or 'limited' because one's senses represent perceptions only, and not external, independent realities. In Kant's perspective on morality, the human will is confronted by a conflict between reason, which appeals to rationality, and desires, which push for the satisfaction of needs. In the face of such a conflict, Kant argued, moral law commands us to act rationally. In other words, he

equated acting morally with behaving in a rational manner, all the while being aware of the finiteness and limitations of intelligence.

From the early 20th century on, one can observe a gradual distancing of 'finite intelligence' from its religious overtones, as it refers more and more to the limitations of human cognition in a naturalistic way. A few earlier sources begin to use the expression in a way approaching the common sense usage of 'limited intelligence', notably Edgeworth (1890: 467), who, together with Henry Carter Adams (1886: 103), who employs 'limited intelligence', is the first economist in the corpus.[9] It is in this period that semantic connections to 'limited intelligence' can be observed. Schiller (1906: 482) for example talks about the "limitations of a finite intelligence." Similarly, Oakeley (1922: 440) finds in a discussion of the scientific method that "cooperation rather than solitude seems to be the method of science in the strict sense. There is, at least, cooperation in thought, through the endeavor to assume the point of view of the least limited finite intelligence, or the collective mind, … ." The clearest semantic connection is due to Smith (1920: 338): "[Tyndall] did not believe that the ultimate truths of the universe could be expressed in words, or that our limited or finite intelligence could as yet comprehend them."

The semantic field from which BR emerges subsequently broadens to incorporate variations of 'rationality', a development that will find its most marked expression in the first occurrence and further diffusion of the notion of BR itself.[10] Rationality first shows up in terms of 'incomplete rationality' as an outgrowth of 'finite intelligence' (Oakeley 1922).[11] Oakeley's article, entitled "On the Meaning of Value," illustrates well the typical gist of philosophical discussion of the concept of finite intelligence in the decades around the turn of the century, while providing a good snap-shot of semantic links from 'finite intelligence' to 'limited' and 'bounded' rationality. Contrary to what its title suggests to the reader who approaches it from the discipline of economics, the article grapples with the old question of how idealism seeks to reconcile the "absolute Infinity of Mind" (433), a notion she takes from Spinoza, who equates it with God, with its individualisation into separate selfs, with the result that "Universal Mind … becomes finite and thus restricted to a single point of

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[9] From the turn of the century, the expression can also be observed in other social sciences, notably sociology (Small 1908: 6; Ford 1909: 99).

[10] The documentation of semantic fields treats all included concepts symmetrically in this regard. No separate argument is thus necessary at this stage to illuminate the transition to another noun (ontologically speaking) or our grounds of including it (exegetically speaking). The only other alternatives, apart from the traditional intuitive approach of the history of ideas, are content-analytical approaches such as the identification of clusters of co-words in the corpus. We do not wish to argue for a privileged position of either the heuristic approach of this paper or the content-analytical approach. We regard both as valid alternatives with both strengths and weaknesses. Ideally, they should complement each other. In the present context, due to constraints inherent to our empirical data, a heuristic analysis is more readily applied to what we have called the 'semantic field' of BR (due to low keyword frequencies), while content-analysis is more readily applicable to the growth of BR itself (due to its explosive growth in the late 20th century, which would render the exegetical tracing of connections an enormous task).

[11] Hilda D. Oakeley (1867-1950), Philosopher, Vice-Principal of London's King's College for Women (Oakeley 1939; Dyhouse 1995).

view in reality" (434).

Ours is a world of "finite intelligences" (439) or "personal beings limited in knowledge and range of experience" (ibid). For these beings of "finite knowledge" (442) to fully comprehend ideals of value is beyond "those logical relations accessible to finite intelligence" (ibid), although they may become partly 'manifest' to practical — as opposed to pure — reason. The differences that result from this only partial apprehension of reality (i.e. the world of ideas) produce an "infinity in the kind and number of the experiences of conscious existence which seems incapable of reduction to the orderly and harmonious system of a fully rational universe" (434-35), a perturbation that Oakeley refers to as the "incomplete rationality" of the universe (ibid). In the idealist context of the discussion, this means that incomplete rationality is a central property of the finite intelligence with limited knowledge. One finds thus many dimensions of the semantic field of BR in Oakeley's paper, whose ultimate interest, one should add, lies in exploring the values guiding human action and interaction. It points thus towards the further development of the semantic field of BR in terms of 'limited rationality'.

'Limited rationality' joins the field in 1945. In a number of ways, and distinguished from the ephemeral use of 'incomplete rationality' by Oakeley, this may be regarded as the beginning of a structural shift in the field, which increasingly realises the communicative potential of various notions of 'rationality'.[12] Compared to the three terms considered so far, it follows a rather different diffusion pattern that exhibits strong exponential growth up to the present. Let us start with the *OED* again, which reveals that 'rationality' as such dates from the 17th century, too, but occurs in the corpus as 'limited rationality' only with Almond (1945),[13] even though 'limited' was *prima facie* lexically available as long as 'rational'. It next appears in Simon (1955).

Herbert Simon first discusses 'limits to rationality' (Simon 1947)[14] in a book and only later mentions 'limited rationality' (Simon 1955) in an article. It appears next in an article by Oliver Williamson (1967) and one by John Harsanyi (1969).[15] While the concept does not occur in full article-length contributions in the intervening period, it seems to have been an important vehicle for the reception of Simon's work (Washington Bell, Weintraub, Earley, and Meyer 1957: 336; Luce 1957: 85). Whereas Simon and Harsanyi make no reference to 'bounded rationality' in their articles,

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[12] In a more comprehensive analysis also incorporating the emergence of the semantic field around rationality, in addition to the field emerging from the notion of intelligence, the immediate post-war period would mark the joining of the two fields.

[13] Almond (1945) is a contribution to political science that makes no reference to Herbert Simon's insights.

[14] See, e.g., Simon (1947: 39-41, 80-84). Here, Simon talks about the limits to rationality and uses 'bound' as a verb: "Perhaps this triangle of limits does not completely bound the area of rationality…" (41). Note that Simon (1947) is not included in our database. References from items in our corpus to this manuscript led us there. Therefore, it is only marginally included in our narrative and excluded from our quantitative evaluations.

[15] It also appeared in a contribution to literary criticism in 1955 (Anderson 1955) that falls outside of our narrative. Harsanyi (1965) uses 'limited rationality' earlier in a book review.

Williamson does employ both concepts interchangeably.

In political science, 'limited rationality' comes to be associated with the insights of Charles Lindblom (1959), which are similar to those of Simon. Also note that Lindblom makes no mention of 'bounded rationality'.[16] His article was voted among the top three policy pieces of all time by readers of *Policy Currents*, the newsletter of the public policy section of the American Political Science Association. This is a possible explanation for the observation that 'limited rationality' shows tentative beginnings from 1945 onwards, but then develops an independent diffusion history shortly after 'bounded rationality' exhibits significant growth rates.

Finally, 'approximate rationality' first appears in our corpus in 1948 in a presidential address to the Canadian Political Science Association (Bladen 1948), in a reference made to a footnote in Frank Knight's (1921) *Risk, Uncertainty, and Profit*: "but the results approximate rationality fairly well on average" (p. 67n). As this reference indicates, it is important to scrutinize the individual occurrences, as we have done for this paper, for here the reference is to the verb and noun 'to approximate rationality', which have quite an opposite interpretation from the adjective and noun 'approximate rationality'. We will return to the concept 'approximate rationality' when we consider Simon's contributions in the following section, after summarizing this one.

The BR field prior to the emergence of BR itself is thus made up of a set of expressions that cluster initially around various restrictions and qualifications of 'intelligence'. 'Finite intelligence' is an expression that is initially mostly used in a technical sense in the philosophical literature, but subsequently broadens into a concept used in a more heterogeneous way similar to 'limited intelligence'. After the Second World War, several expressions related to 'rationality' joined this field within little more than a decade, triggering a process in which one concept stopped being used in favour of another. A pivotal text marking the transition could be identified in Oakley's (1922) use of 'incomplete rationality'. Whereas the discourse community gradually ceased using 'finite intelligence', 'limited rationality' saw a conceptual renaissance in the 1980s, when the hitherto marginal expression started being used more frequently.

4. The Emergence of Bounded Rationality

In the previous section, several changes in the semantic field of BR have been identified and described in detail. None of these alone can account for the emergence of BR, which necessarily involves an innovative event, as discussed in this section. In particular, we will suggest that Herbert Simon, almost certainly the first person to have used the concept 'bounded rationality', tentatively applied the first best expression that seemed available on the basis of associations with what he

[16] Interestingly, Lindblom also does not mention 'bounded rationality' explicitly in a book published a few decades later (Holmes 1979).

sought to express. In subsequent writings, he consciously refined and replaced original concepts such as ‘approximate rationality’ and ‘limited rationality’ until setting for ‘bounded rationality’ as a stable term. As Simon (1999: 23) notes, he began to use this concept after a while: “You have to realize about the bounded rationality terminology that I began to use this as a label for the things that economists needed to pay attention to — and were not. It was never intended as a theory in any sense”. Whereas Simon sought to construct a new label with a new meaning, this concept was embedded in what we call a semantic field. In Simon’s (1957a: 63) words: “Like Humpty Dumpty, we will insist that a word means what we want it to mean. But if our aim is to construct a body of science, and if we already have in view the general range of phenomena to be explained, our decisions may be willful, but they must not be arbitrary”.

The BR concept, in all likelihood, first appeared in print in *Models of Man* (Simon 1957a: 198).[17] In his mature work, Simon used the concept to “designate rational choice that takes into account the cognitive limitations of the decision-maker — limitations of both knowledge and computational capacity” (Simon 1987: 266). Through the use of BR, Simon sought to criticize neoclassical economists for their lack of interest in the formal foundations of rationality. In particular, he disapproved of what he considered to be the four basic assumptions of neoclassical economics. First, the analysis started from the presupposition that each economic agent had a well-defined utility or profit function. Second, the decision-maker knew all alternative strategies. Third, neoclassical economics assumed that all the consequences that follow upon each of these strategies could be determined with certainty. Finally, the comparative evaluation of these sets of consequences was assumed to be driven by a universal desire to maximize expected utility or expected profit.

Instead, Simon advanced the BR concept in an effort to include the whole range of limitations on human knowledge and human computation that prevent economic actors in the real world from behaving in ways that approximate the predictions of neoclassical theory. First, the concept represents the idea that decision-makers were confronted by the need to optimise several, sometimes competing, goals. Second, instead of assuming a fixed set of alternatives among which the decision-maker chose, Simon postulated a process for generating alternatives. Third, he argued that individuals had difficulty coming up with original solutions to problems. Finally, instead of assuming the maximization of a utility function, Simon’s BR concept included a satisficing strategy.

Focusing on external, social constraints and internal, cognitive limitations to decision-making, Simon stressed the process rather than the outcome of decision-making. Awareness of these bounds,

[17] Claims to first occurrences in vast historical corpus such as ours can only ever be tentative, forming a hypothesis inviting our colleagues in the history of economics and related fields to put it to the test and, if possible, refute it.

according to Simon, caused people to use heuristics and satisfice. Loosely articulated heuristics, or rules of thumb, Simon argued, governed the process of gathering information and choosing alternatives. According to Simon, these heuristics were employed generally because they had been proven successful in the past. Furthermore, they implied that the decision-maker was searching merely for an adequate solution. That is, people satisficed, they accepted the first solution that was satisfactory according to a set of minimal criteria. These aspirations were adjusted in the interaction among heuristics and satisficing. For Simon, BR and the associated insights concerning aspiration levels, heuristics, and satisficing were thus central to his contributions to so-called behavioural economics. The concepts Simon developed in BR served as a springboard to his interpretation of artificial intelligence. The same ideas of 'heuristic' or 'rule-bound' search, 'satisficing' behaviour, and 'goal, subgoal' strategy that shaped Simon's theory of BR also became key concepts in his problem-space approach to reproducing human-style reasoning.

Once we attend to the emergence of BR in Simon's work, we find that it was well embedded in what we refer to in the present article as the BR field. In a number of ways, BR became the new core expression of the BR field that previously relied on the 'finite' - 'limited intelligence' pair, due to a gradual process of switching from one concept to another. As we have seen above, experimentation with restricted notions of rationality began well before Simon coined the term 'bounded rationality' in 1957. Figure 2 revisits the development of the BR field from 1945 onwards, but this time including all relevant series (apart from 'finite' and 'limited intelligence' that have been left out for reasons of transparency; cf. Figure 1).

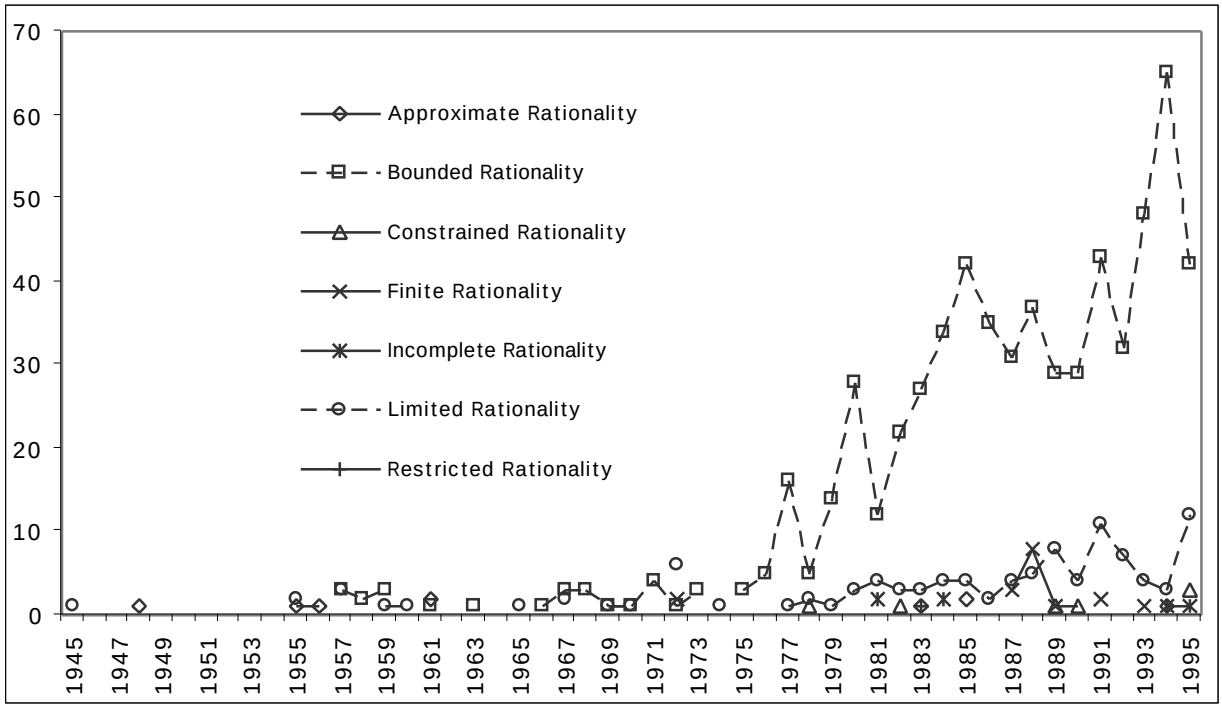


Figure 2: The Semantic Field including BR (1945-1995, per annum counts)[18]

As can be seen from Figure 2, both 'limited rationality' and 'approximate rationality' predate BR, albeit only by a couple of years. Short of Simon's appeals to 'limits to rationality' in *Administrative Behavior* (1947),[19] 'limited rationality' occurs in the corpus three times before 1957: in the field of politics in common sense usage (Almond 1945; Anderson 1955) and in Simon (1955), used in a technical sense. 'Approximate rationality' occurs in our database twice before 1957, once in a presidential address to a political science association (Bladen 1948) and in Simon (1955).[20] As noted in the previous section, the ephemeral occurrence of 'incomplete rationality' constitutes the first and at least equally significant tentative experiment at the margins of the BR field. Yet, Oakeley's use of 'incomplete rationality' in 1922 and Simon's appeals to 'limited rationality' could not be more different. Oakeley's usage bears all the signs of a serendipitous conceptual innovation that never got 'off the ground'. In fact, it was only taken up again during a brief conceptual renaissance in the 1980s, after other 'rationality' concepts had taken off, most notably BR itself. Hence, one might refer to this as an instance of a serendipitous conceptual invention that failed to transform into an innovation.

Simon, then, did not invent the concept of 'limited rationality'. The way it occurs in Almond (1945) indicates that as a common-sense notion, it must have been well established in political discourse.[21] In his own rational reconstruction, Simon suggests that he picked up the concept 'limited rationality' from John R. Commons and Chester Barnard (March and Simon 1958: 169-171; Simon 1991: 86-87). However, Commons referred to 'limiting factors' (Commons 1934: 58, 90) and Barnard spoke of 'strategic factors' (Barnard 1938). *Administrative Behavior* talks of 'limits to rationality' instead. Here, Simon employs this concept to establish a contrast between 'administrative man' and 'economic man'. In our corpus, Simon first appeals to 'limited rationality' in 1955. Here, he uses it to distinguish models of 'limited' rationality and those of relatively 'global' rationality (Simon 1955: 112-113). In this paper, he also effects a conceptual translation by turning the expression into a technical term requiring a model. In fact, of all of Simon's research,

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[18] This figure has not been adjusted for the total number of entries in JSTOR, since these numbers exhibit a roughly linear trend that does not affect the graph too much for our purposes. A normalized figure is available from the authors by request.

[19] Note that Simon (1947) is not included in our database. References from items in our corpus to this piece led us there. Therefore, it is only marginally included in our narrative and excluded from our quantitative evaluations.

[20] 'Approximate rationality' also appears in Simon (1956), the companion paper to Simon (1955). Note that the companion paper is not included in our database. References from items in our corpus to this piece led us there. Therefore, it is only marginally included in our narrative and excluded from our quantitative evaluations.

[21] We might even speculate about the existence of a semantic link, possibly indirect, between 'incomplete rationality' as documented in Oakeley (1922), and the emergence of 'limited rationality' as a common-sense term, both developments being traces of a secularising move from the old contexts of 'finite intelligence' on to the new grounds opened up by expressions related to 'rationality'.

this paper is the one that comes closest to the mathematical format with which economists are comfortable and that economists who wish to refer to bounded rationality most commonly choose for citation (Simon 1991: 165). Perhaps not surprisingly, very few economists refer to the less mathematical companion paper 'Rational Choice and the Structure of the Environment' (Simon 1956).[22] Simon (1955) and its companion paper (Simon 1956) both appeal to 'approximate rationality' as well. Again, the focus is on developing a technical interpretation of 'administrative man' as an alternative to 'economic man': "The broader aim, however, in constructing these definitions of 'approximate' rationality is to provide some materials for the construction of a theory of the behavior of a human individual or of groups of individuals who are making decisions in an organizational context" (Simon 1955: 114).

Simon experimented this with several expressions derived from the field. During the transition from 'approximate rationality' and 'limited rationality' to 'bounded rationality', Simon discusses 'boundary conditions' in his early work (e.g. Simon 1943). In Simon's (1991) words: "By *boundary conditions* I mean the assumptions that have to be made" (83, original emphasis). In other earlier publications (e.g. March and Simon 1958: 169-71; Simon 1955) there is also mention of the 'boundaries of rationality' or 'the boundary'. Orthodox economics, according to Simon, posits people who behave rationally subject to a wide range of boundary conditions. In his opinion, the 'action' lies in the boundary conditions. He made this insight clear by combining 'boundary' and 'rationality' into 'bounded rationality'. The concept 'bounded rationality' gradually advances in later editions of *Administrative Behavior*, as evidenced by the added introductions, which highlight that the book is concerned with "the boundary between the rational and the nonrational aspects of human social behavior" (Simon 1957b: xxiv, 1976: xxviii, 1997: 18).[23] Whereas the central concern remains the same throughout the different editions of the book, its description changes, reflecting a gradual transition from the concept of limited rationality to that of BR:

- "[H]uman behavior is intendedly rational, but only *limitedly* so" (Simon 1957b: xxiv, our emphasis).
- "[H]uman behavior is intendedly rational, but only *limited* so" (Simon 1976: xxviii, our emphasis).
- "[H]uman behavior is intendedly rational, but only *boundedly* so" (Simon 1997: 88, our emphasis).

[22] Whereas the first paper inquires "into the properties of the choosing organism," the sequel inquires "into the environment of choice" (Simon 1955: 100). Furthermore, the sequel "casts serious doubt on the usefulness of current economic and statistical theories of rational behavior as bases for explaining the characteristics of human and other organismic rationality. It suggests an alternative approach to the description of rational behavior that is more closely related to psychological theories of perception and cognition…." (Simon 1956b: 138).

[23] To clarify, these comments are absent from the first edition (Simon 1947).

Our evaluation of the transition to BR returns us to the publication likely to be the first to mention the BR concept. In our database, BR first appears in book reviews of *Models of Man* (Simon 1957a).[24] The volume contains sixteen reprinted articles, and first presents BR as a principle in the introduction to the final section of contributions on 'Rationality and Administrative Decision Making': "The alternative approach employed in these papers is based on what I shall call the *principle of bounded rationality: The capacity of the human mind for formulating and solving complex problems is very small compared with the size of the problems whose solution is required for objectively rational behavior in the real world—or even for a reasonable approximation to such objective rationality*" (198, original emphasis).[25] Witnessing the transitional phase between 'limited rationality' and BR, several of the book reviews refer to both concepts. In *Science*, Luce (1957) sees 'limited rationality' as one of the three theses of the book (85) and views the principle of BR as a component of this thesis (85). Mosteller (1959), in sociology, refers to 'limited or bounded rationality' and links it with Simon's later work on artificial intelligence. Solow's review (1958) appears in an economics journal and notes the principle of BR as a reasonable insight, though it suggests that the characterization of 'economic man' is not entirely fair (83). Finally, in mathematics, Good (1958) picks up on the principle of BR, but makes no mention of 'limited rationality', just like Solow.

In contrast to all other expressions, then, BR exhibits a significant amount of conceptual experimentation at the hands of Simon, who drew from the field initially when he employed 'limited' and 'approximate' rationality. But one may speculate that for his purpose of founding a whole research programme in economics on a new concept of rationality, 'limited rationality' suffered from the negative connotations of 'limited intelligence' mentioned above, while 'approximate rationality' did not exhibit the linguistic 'robustness' of 'bounded rationality', which managed to tie rationality to the important technical implications of boundary conditions. Further evidence for Simon's attempts to strengthen the concept is due to the 'principle' label he attached to it.[26] One is tempted to say that history proved Simon's conceptual experimentation right, although a richer analysis is necessary to substantiate this point in more detail. Of all the expressions of the semantic field, only BR mustered an explosive diffusion record. This brings us to a closer look at

[24] It also shows up in book reviews of *Organizations* (e.g. Kaufman 1959), with most mentioning 'limited rationality' as well, as noted before. It also appears in a book review of James March's *Handbook of Organizations* (March 1965) by McWhinney (1966), who makes no mention of 'limited rationality'.

[25] The book has an index entry for 'bounded rationality, principle of' pointing to the introduction to Part IV that explicitly discusses this 'principle' but also to papers that do not do so.

[26] As an aside, in modelling BR, Simon employed engineering mathematics. Simon (1991), whose father had been an engineer, noted in his autobiography: "It gradually dawned on me that the path I was following in my professional maze was returning me to the parental calling, and not only because I had chosen to teach at an engineering school. … I decided I had been a closet engineer since the beginning of my career" (108-109).

the initial diffusion of BR and a bird's eye view of the subsequent dispersion.

5. The Consolidation and Diffusion of Bounded Rationality

Following *Models of Man* (Simon 1957a), at first our database entries frequently refer to BR as a 'principle' (e.g. Mitchell 1961; Laffont 1975). In fact, most early appeals to the concept include this label. However, when they discuss the interpretation of this principle, they oftentimes use 'limited rationality', which shows that the latter was current usage in the 1940s and 1950s. Yet, a transition occurred when this normal usage was translated into a 'principle', exchanging 'limited' with 'bounded'. In other words, the concept of BR becomes special only in the course of institutionalisation, something of which the first users were unaware. This strengthens our assessment of the innovation as an instance of conceptual engineering. Following the second edition of *Administrative Behavior* (Simon 1947; 1957b), the next label that is widely used is 'theory' (e.g. Harsanyi 1969; Kunruether and Slovic 1978). Following *Models of Bounded Rationality* (Simon 1982), appeals to the 'model' of BR become widespread later on (e.g. Chenault and Flueckiger 1983; Moe 1984; Chayes and Handler Chayes 1993).[27] Oliver Williamson appears to favour BR as an 'assumption' (e.g. Williamson 1981a, 1981b) and other less frequently used labels are 'approach' (e.g. Calvert 1985; Rosenthal 1989), 'concept' (e.g. Loasby 1971; Olson, White, and Shefrin 1979), 'condition' (e.g. Rycroft and Szyliowicz 1980; Silverberg, Dosi, and Orsenigo 1988), 'elements' (e.g. Leibenstein 1979; Rubinstein 1993), 'framework' (e.g. Simon 1979; Bray and Savin 1986), 'heading' (e.g. Tversky and Kahneman 1981; Hirshleifer 1985), 'hypothesis' (e.g. Appelbaum and Harris 1977; Radner 1992), 'idea' (e.g. Simon 1979; Evans 1981), 'issue' (e.g. Hart and Moore 1988; Abreu and Matsushima 1992; Petersen 1993), 'notion' (e.g. Eulau 1959; Williamson 1967), 'paradigm' (e.g. Padgett 1980), 'problem' (e.g. Smale 1980; Alexander and Fennell 1986; Lazerson 1988), 'tradition' (e.g. Padgett 1981; Cohen and Axelrod 1984), and 'view' (e.g. Russell and Thaler 1985; Lau, Smith, and Fiske 1991; Sebenius 1992). Not until the 1970s do we witness 'boundedly rational' in our database (e.g. Futia 1974; Appelbaum and Harris 1977; Schotter 1979). After its initial appearance, it follows the pattern of the noun. In our view, this is further indication of the successful diffusion of BR, as a technically coined noun transmutes into adjective usage.

The first appearance in an article occurred in the field of political science (Eulau, Wahlke, Buchanan, and Ferguson 1959), as did its second one in an article (Mitchell 1961). In fact, most of the important shifts in the semantic field seem to have originated from the discipline of politics, beginning with Almond (1945) and leading up, via the book reviews, to the first articles that take up

[27] Simon (1955) refers to models of 'limited' rationality.

the concept. Figure 3 provides an overview of the distribution of fields for BR entries in our database, leaving out those with relatively few entries.

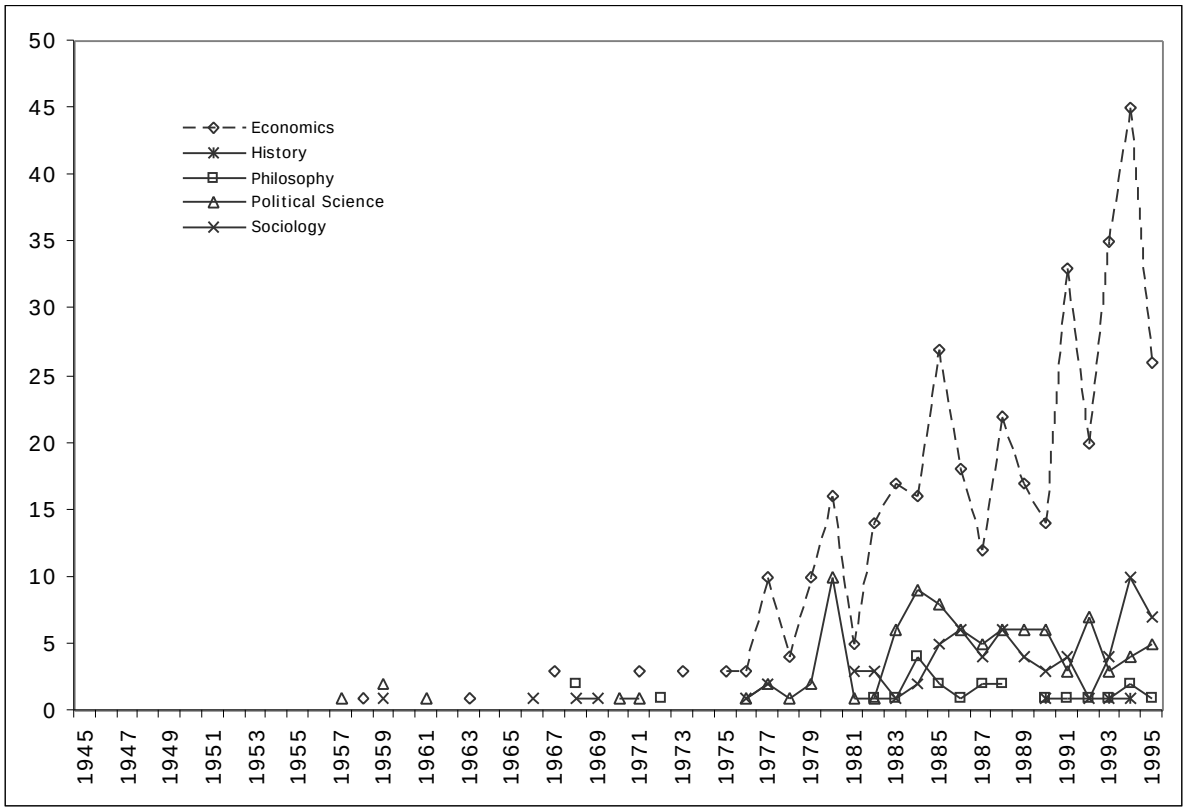


Figure 3: The Field Distribution for BR Entries (1945-1995, per annum counts)[28]

As suggested before, the likely explanation for the persistence of political science entries is the use of 'limited rationality' by Charles Lindblom (1959).

Returning to economics, the first articles in English to refer to the concept appeared in 1967 (Day 1967; Williamson 1967). In our database, the first article by Simon to contain BR was not published until 1978 (Simon 1978). Not included in our corpus is the first appearance of BR in an article title (Simon 1972). By the later 1980s, 'bounded rationality' was firmly entrenched as one of the core concepts of economics, documented by its appearance in the main professional dictionary of the discipline of economics (Simon 1987). At about the same time, one can witness its increasing appearance in book titles, pioneered by Simon himself (1982), and followed by several others a decade later (Egidi, Marris, Simon, and Viale 1992; Sargent 1993; Rubinstein 1998; Gigerenzer

[28] This figure has not been adjusted for the total number of entries in JSTOR, since these numbers exhibit a roughly linear trend that does not affect the graph too much for our purposes. A normalized figure is available from the authors by request.

2001).[29]

The qualitative shifts discussed so far, indicated by the changes in the conceptual modalities (Klaes 2001) in which the term manifests itself in the literature, such as its appearance in quotation marks, titles, etc., throw an important light on the changing status of the concept in the various literatures in which it emerged, and thus portray a process of institutional change. This brings us to a brief overview of attempts to influence the momentum of this growth process on the part of a number of prominent branches of contemporary economics, such as transaction cost economics, game theory, and rational expectations economics.

Simon's behavioural insights stand in sharp contrast with the more recent contributions to this field. Starting from empirical and experimental counterevidence to the strong rationality assumptions employed in neoclassical economics, these formalize and test psychological predictions. After identifying ways in which behaviour differs from the neoclassical model, they show how alternative theories can explain apparent anomalies. Along the way, they suggest that market forces such as competition and arbitrage, learning on the part of decision makers, and some sort of evolutionary mechanism cannot eliminate the importance of their insights. This more recent research within the field goes back to the work of Amos Tversky, Daniel Kahneman, Paul Slovic, and others (e.g. Kahneman and Tversky 1974, 1979; Kahneman, Slovic, and Tversky 1982; Tversky and Kahneman 1987). Starting from the perspective of utility-maximization and Bayesian probability judgments, Kahneman, Tversky, and their followers evaluated the cognitive character of conformity or deviation from these benchmarks. Their contributions may be divided into three areas: heuristics and biases, framing effects, and prospect theory. According to Colin Camerer (1999), "this sort of psychology provided a way to model bounded rationality which is more like standard economics than the more radical departure that Simon had in mind. Much of behavioural economics consists of trying to incorporate this kind of psychology into economics."

The broadening of the notion of BR is further witnessed by the contributions of Oliver Williamson (e.g. 1975) and of Richard Nelson and Sidney Winter (e.g. 1982). First, Williamson is responsible for the prominent role played by BR in transaction cost economics. Williamson sought to link the idea of conflict of interest with the idea of information limitations and saw organizational forms as implicit or explicit solutions to the problems of decision and control created by opportunism and BR. Opportunism refers to the fact that there is conflict of interest within, as well as between, organizations, and that participants in an organization will lie, cheat, and steal in their own self-interest if they can. BR makes complete contracting infeasible because not everything can

[29] According to the *Dissertation Abstracts* database available via http://www.umi.com, the first doctoral dissertation with the concept in its title was by Steve Fuller (1985) but it should be noted that eleven years earlier, Carl Futia (1974) published a dissertation with 'boundedly rational' in the title.

be known and there are limits to the capabilities of decision makers for dealing with information and anticipating the future. However, Williamson was reluctant to accept the notion of satisficing, primarily because he thought it would denote irrational behaviour. At the same time, Simon himself considered satisficing to be a close companion of the concept of BR. Second, Nelson and Winter offered an alternative interpretation of BR with their research on an evolutionary theory of business firm growth. It emphasizes differential survival as a primary basis for changing populations of firms, and sees firms as being selected upon by virtue of their fit to the environment. Nelson and Winter stressed the inability of firms to carry out the necessary calculations for optimisation, because the firm will not 'know' all the things of which it is capable, because all future contingencies cannot be foreseen, because mistakes can be made, and so forth. In Nelson and Winter's view, the notion of satisficing can account for persistence of routines in their evolutionary theory. At the same time, they resisted incorporating organizational motives, an insight considered to be central by Simon.

More recently, BR has appeared in game theory as well as macroeconomics. First, game theorists have tried to capture BR by replacing rational players with computing devices such as Turing machines, finite automata, or neural network algorithms (e.g. Aumann 1981; Radner 1980). Yet, whereas Simon had considered the concept of BR to be an alternative to neoclassical economics, game theorists have used it in an attempt to strengthen it. For them, modelling players as being boundedly rational permits solutions to be reached that they want to obtain but cannot do so from fully rational players. In particular, the concept has been used in the area of refinements of the Nash equilibrium, for the study of the applicability of the Folk Theorem, and on the problem of equilibrium selection. Second, much like game theorists, macroeconomists have sought to incorporate the concept of BR in an effort to strengthen their mainstream insights (e.g. Bray and Kreps 1987; Sargent 1993). Sargent, for instance, has sought to reinforce rational expectations by focusing on convergence to this equilibrium through boundedly rational learning. He also tried to use the concept of BR to deal with some of the problems associated with rational expectations, such as multiple equilibria and the computation of equilibria.

A final indicator of the stable institutionalisation of BR by the 1990s can be found in the fact that authors, including Simon himself, tend to read it anachronistically into *Administrative Behavior*. Even though the first edition does not include the BR concept, Simon (1991) writes in his autobiography, when discussing the book: "[T]he idea of bounded rationality … appears to be the most novel and original component of the work" (87).[30] Others also attribute the concept to the first

[30] Simon (1991) further notes that *Administrative Behavior* "was built around two interrelated ideas that have been at the core of my whole intellectual activity: (1) human beings are able to achieve only a very bounded rationality, and (2) as one consequence of their cognitive limitations, they are prone to identify with subgoals" (88).

edition of *Administrative Behavior*. For instance, whereas Moe (1984)[31] and Bendor and Hammond (1992)[32] note that it was developed in the book, Ouchi and Wilkins (1985)[33] and Beck, Rainey, and Traut (1990)[34] do the same while mistakenly dating the manuscript in 1945. And despite the fact that Simon's 'A Behavioral Model of Rational Choice' (Simon 1955) talks of 'boundaries' without containing 'bounded rationality' as a concept, Simon (1991) notes in his autobiography, when covering the article: "Of all my writings on this topic, the 'Behavioral Model' paper comes closest to the mathematical formal with which economists are comfortable. Hence, economists who wish to refer to bounded rationality … most commonly choose this paper for citation" (165). Others also make the same attribution (Tversky and Kahneman 1981; Schoemaker 1982; Steiner 1983; Hirshleifer 1985).[35]

6. Conclusion

Starting with the occurrence of 'limited intelligence' in 1840 and that of 'finite intelligence' in 1880 and through the appearance of 'incomplete', 'limited', 'approximate', and 'administrative' rationality during the first half of the previous century, we have developed a perspective on the appearance of the concept of 'bounded rationality' that interprets this emergence in the context of the semantic field of BR. We view our study as a contribution to an institutional history of the concept of bounded rationality, which further attends to the interactive and open-ended linkages, networks, and processes in which 'bounded rationality' is embedded.

During our heuristic search, we learned that both 'finite intelligence' and 'approximate rationality' suffered through a process of conceptual fading, in which the expressions gradually ceased to be used. 'Limited intelligence' exhibited a conceptual renaissance when the hitherto marginal expression acquired a specific interpretation. Finally, we witnessed conceptual switching

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[31] See Moe (1984: 743): "At the heart of Simon's contribution is his model of bounded rationality, first developed and applied to organizations in *Administrative Behavior* (1947)."

[32] See Bendor and Hammond (1992: 313): "In 'Administrative Behavior', Simon [1947] began with the premise that individuals are boundedly rational."

[33] See Ouchi and Wilkins (1985:464-65): "The legitimation for combining a belief in organizational rationality with the empirical observation of organizational nonrationality was offered by Herbert Simon in 1945. … With this idea of 'bounded rationality', Simon provided the basis for the coupling of the rational and the nonrational views of organizations."

[34] See Beck, Rainey, and Traut (1990: 72): "The work of Herbert Simon (1945) has long since established that decision making in organizations normally takes place under conditions of limited cognitive capacity and information, such that the intended but bounded rationality of decision makes leads to satisficing rather than optimising."

[35] See Kahneman and Tversky (1981: 458): "[I]ntellectual limitations [are] discussed by Simon [1955] under the heading of 'bounded rationality'." See Schoemaker (1982: 545): "The bounded rationality view (Simon, 1955) of humans is that of an information processing system." See Steiner (1983: 376): "[T]he comprehensive nature of human reason has been challenged by the idea of bounded rationality [Simon (1955)]." See Hirshleifer (1985:61): "Under the heading of 'bounded rationality', Herbert A. Simon (1955, 1959) has contended that a person faced with a complex mental task will not attempt to strictly maximize, but will be content instead merely to 'satisfice'."

from ‘intelligence’ to ‘rationality’ from the nineteenth to the twentieth century, during which the discourse community stopped using one concept in favour of another.

We established the pivotal position of the paper by Hilda Oakeley (1922).[36] Next, we presented Simon’s development of ‘bounded rationality’ as the result of conscious experimentation with competing expressions drawn from the semantic field, which finally crystallised in a focal expression. In the process, he appealed to ‘limited rationality’ and ‘approximate rationality’, which had both earlier occurred in political discourse, before settling on ‘bounded rationality’. While refining and replacing the original expressions, Simon tentatively connected ‘boundary’ and ‘rationality’ into the concept that has been the main focus of our paper. His endeavours were strengthened by his appeals to the concept as a principle. Finally, we gave an overview of the subsequent diffusion of the concept in which others translated ‘bounded rationality’ in a new way and started using it in their writings. In the process of institutional change, the status of the concept changes and efforts emerge to influence the momentum of its growth process.[37]

Contrary to some existing histories of the concept of bounded rationality, we have explored a very long time horizon, going back to 1840. Considering antecedent notions that were floating around long before Simon coined the phrase enabled us to do this. In opposition to many existing uses of bounded rationality, we have stressed the ambiguity that still surrounds it today. At variance with certain existing histories of the concept, we have refrained from suggesting that Simon has exclusive rights on how to understand and use the concept correctly.

What we find remarkable about Thomas Sargent’s appeals to bounded rationality is that he apparently felt obliged to connect the concept with Herbert Simon’s contributions. In response, conceptual history is interested in tracing words rather than people and their ideas. What we find fascinating about Robert Aumann’s observations is that he seemingly had little interest in its earlier engineering, but sought to link bounded rationality with Roy Radner’s insights. In response, conceptual history focuses on the evolving heterogeneity of the expression. What we find intriguing about John Conlisk’s survey is that he presumably felt no compulsion to cover the content of the concept. In response, conceptual history traces the semantic field from which the concept emerged. As such, it uncovers linkages, networks, and processes that might otherwise escape the attention of history of economics.

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[36] Some historians of economics might be tempted to suggest that many ideas of BR analysis can be found in Oakeley’s paper, albeit transcribed into a different domain. However, *Begriffsgeschichte* is not interested in tracing precursors of ideas, with no anachronism or rational reconstruction involved.

[37] We hesitate to speculate that the emergence of BR as the prevalent expression in the field is due to Simon’s ascendancy, Williamson’s adoption of the concept, and subsequent bandwagon effects, for this would require additional social history evaluations that fall outside the scope of our paper.

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